

Business Research Proposal

Title of the Proposal

An AI-Powered Framework for Real-Time Marketing Strategy
Optimization in Retail & E-Commerce

Business Research Proposal Competition

Theme: AI for Business Transformation

Track:

AI in Strategic Business Decision Making

Team Details

Team Name: Idea Igniters

Team Members:

Name – Neel Javiya (AI & DS)

Name – Rahul Mandal (AI & DS)

Name – Vicky Singh (AI & DS)

Name – Nakul Jadhav (AI & DS)

Name – Prarthi Patel (AI & DS)

Institute / College Name: Adani Institute of Digital Technology
Management [AIDTM]

Submitted To:

Adani Institute of Digital Technology Management
Gandhinagar, Gujarat

TABLE OF CONTENTS

Sr No.	Title	Page No.
1	Executive Summary	1
2	Background & Industry Context	1
3	Problem Statement	3
4	Existing Solutions Review	4
5	Research Gap	5
6	Proposed AI Solution	6
7	Research Objectives / Hypothesis	7
8	Methodology (How the solution will be tested)	8
9	Expected Outcomes	10
10	Business Value	11
11	Implementation Plan	12
12	Risks & Ethics	13
13	Conclusion	15
14	References	16

1. Executive Summary

Retailers and online shops today collect more customer data than ever before - but the majority of them do not know how to use it effectively. Marketing campaigns are still built off of large customer groups and outdated data, meaning that businesses are wasting money on ads and promotions that are not reaching the right people at the right time. This is a serious and growing problem: bad marketing decisions cost U.S. retailers alone more than \$756 billion in lost sales every year.

We're looking at the Retail and E-Commerce industry - one of the fastest- and most competitive industries out there - and there is a desperate need for smarter and faster marketing decisions.

We propose SMART - AI (Strategic Marketing Adaptive Real - time AI) a new Artificial intelligence powered system that pulls together customer data from all channels - website, email, social media, in store - and uses machine learning to predict what customers want and automatically make the best marketing decisions in real time.

The primary research goal is to determine whether SMART-AI indeed is better for marketing than traditional methods. We anticipate that retailers who use SMART-AI will attract far more customers to purchase, waste less money promoting failed ads and lose fewer customers to the competition.

If successful, SMART - AI could help retailers drive millions more in added sales, reduce marketing waste by up to 40% and build more robust, longer lasting relationships with their customers - with clear and measurable business value.

2. Background & Industry Context

2.1 Introduction to the Industry

Did You Know retail and E-Commerce is one of the biggest, fastest growing sectors out there?

It includes everything from the tiny online shops to the huge like Amazon and Alibaba.

According to Statista, in 2024 the global e-comm market reached \$6.3 trillion and it's expected to exceed \$8 trillion in 2027 (Statista, 2025).

This industry includes a vast range of products - from clothes to electronics, groceries to home goods, and reaches billions of customers through websites, mobile apps, brick and mortar stores and social media.

2.2 Current Trends and Challenges

Online shopping has massively increased in the past few years. And by 2027, people worldwide are projected to spend over \$8 trillion shopping online - and that's crazy.

Because of this growth, the retailers, be it through online retailers or physical stores, are under a lot of pressure to stand out and provide a better experience for customers.

The problem is that most retailers are still working with old fashioned marketing methods. They place customers in broad groups (such as "young women" or "budget shoppers") and send everyone in that group the same message. But that approach does not work very well anymore because every customer is different - you know - with different tastes, different shopping habits, they react differently to advertisements and promotions.

This is where Artificial Intelligence (AI) comes in. AI is a type of computer technology that is able to look at huge amounts of data very quickly and find patterns that humans would miss. In the context of marketing, AI can examine how a customer sets foot on a website, what their purchases are, what they do not purchase, and what they return - and use all that information to determine exactly what that customer wants, and when they want it.

2.3 Key Industry Facts

Current Challenges in Retail	What AI Can Do
Online shopping market: \$6.3 trillion in 2024	Cost of getting new customers has risen 60% in 5 years
8 out of 10 customers prefer personalized offers	Personalized emails get 6x more clicks
AI in marketing is growing to \$107.5B by 2028	AI can cut wasted marketing spend by up to 37%
Shops using AI see 20-30% more sales	Real-time personalization can give 200-500% return
70% of online shopping carts are abandoned	Companies using data grow 23x faster
Only 22% of retailers currently use advanced AI	87% of marketers plan to spend more on AI

Even though the opportunity is clear, most retail businesses haven't really figured out how to use AI in their marketing properly. They've got loads of customer data but not idea how to make it into action-able decisions. That is the very issue that this research aims to solve.

2.4 Why This Problem Matters Right Now

This issue is to super important right now for three big reasons. First, online shopping exploded after the pandemic caused by Covid-19 and we are now generating way more customer data every day - but most of this data just gets wasted. Second, today's customers have more choices than ever before, so if a brand sends them an irrelevating message they're going to just hop to a competitor. Third, the cost of acquiring new customers has jumped 60% over the last five years, which means that we need to keep current customers happy with smart, personalized marketing more than ever - and it's actually more profitable. Retailers that fail to address this issue in the near future are at risk of never catching up with the competition.

3. Problem Statement

The Main Problem

Retail and online shopping businesses collect huge amounts of customer data every day, but they don't really have the right tools or systems to make proper use of that data. As a result, their marketing campaigns end up being pretty ineffective - wasting cash, losing customers and missing out on sales.

Today, the majority of retail marketing has three big problems:

1. Problem 1 : Always Playing Catch-Up Most of the marketing decisions are made with data that is often a few weeks or even months out of date. By the time a retailer finally does something, customer preferences change, so they're trying to get the right message at the wrong time.
2. Problem 2 : One-Size-Fits-All Marketing Retailers place millions of different customers into a few, broad categories. But real customers are way more complicated than that and that makes most marketing feel generic and out of touch to the person getting it.

3. Problem 3 included: "Data Stored in Different Places: A customer's info might live in one system for online shopping, another for in-store purchases, and yet another for email marketing - these systems don't talk to each other." As a result, retailers never get a complete picture of who their customers are.

These problems are very expensive. Research shows that bad, irrelevant marketing is costing U.S. retailers over \$756 billion in lost sales annually. Retailers that address these problems using AI are leaving their competitors in high gear.

3.1 What Is Affected by This Problem

This problem isn't simply a one off issue for one department - it actually reverberates throughout the entire corporate ecosystem. Marketing teams end up blowing through their budgets by sending the wrong message to the wrong people. Sales teams lose prospects simply because they didn't reach them at the right time. Customer-service crews get hit with an increased volume of complaints from customers who feel that the brand does not get them. At the executive level, leaders are seeing smaller profits and lackluster growth where there should be more. Bottom line, bad marketing hurts everybody from the marketing squad to the individual customer to the company's bottom line.

4. Existing Solutions Review

There's already a mixture of tools and methods that retailers employ to attempt and fix these problems. But to be honest, each one of them has major weaknesses:

4.1 Real-World Examples of Current Approaches

- Amazon's recommendation engine is rather rudimentary - it just recommends products based on what a customer has already purchased or viewed. That's fine to nail product suggestions, but it's flat on bigger decisions, like when to send an email blast or how to spend money on ads, or how to win back a customer about to drift away.
- Zara uses crunching data on sales to determine the products to restock and promote. That's a smart way to use data for inventory purposes, but it's all about managing the stock-not about making the experience more tailored for individual customers across the different marketing channels.
- Netflix knows what shows to recommend to each user, which is impressive, but all goes on inside the Netflix app. Retail brands must coordinate across an entire stack, including websites, emails, social media, paid advertising, and physical stores, simultaneously.

These examples illustrate that although AI is already being implemented in some channels of the retail and e-Commerce world, no company has constructed a single complete system that can manage the entire marketing strategy in all channels simultaneously.

Current Solution	What It Does Well	Key Weakness
Rule-Based Systems (Simple IF-THEN rules)	Easy to set up and cheap to run	Very rigid cannot learn or change on its own when customer behaviour changes
Traditional A/B Testing (Testing two versions of an ad)	A reliable, well-known method	Very slow — tests only one thing at a time and ignores what is happening right now
CRM Customer Grouping (Sorting customers by basic info)	Uses existing customer data that businesses already have	Groups are too broad and miss the fine details of what each customer actually wants
Basic Recommendation Engines (Like Amazon's 'You may also like')	Works well for suggesting products and scales easily	Only works on one channel (e.g., website) and cannot be explained to managers
Third-Party Marketing Software (e.g., Salesforce, Adobe)	Powerful and ready to use out of the box	Very expensive, hard to customize, and not built for any specific business's needs

5. Research Gap

So I've been digging through all these research papers talking about using AI in marketing, like how it will recommend a product or even predict who might cancel

or leave, but still, no one has created one system that takes all those things and puts them together.

The Gap This Research Fills

No system is up to date pulling together customer data from every channel: website, app, email, store, and in real time they predict what customers want and make smart marketing choices, all at once. This research plans to build and test exactly that sort of system.

Picture it like this - most of the AI tools out there are just like individual puzzle pieces: one for email, one for ads, one for product recommendations. What's missing is a way of stitching them all together into a bigger picture. That is what this research is about.

We're planning to build and evaluate a new AI system, SMART-AI (Strategic Marketing Adaptive Real-Time AI) system, that ties all the marketing functions together and helps retailers make smarter choices across the board in real-time.

6. Proposed AI Solution

6.1 What Is SMART-AI?

Hey, so SMART - AI is an abbreviation for Strategic Marketing Adaptive Real - Time AI. Basically, it's this system that pulls together all kinds of AI and machine- learning tech to help retailers get a better handle on their customers and make way smarter marketing calls on autopilot, instantaneously.

In short, SMART - AI is just watching what the shoppers are doing, guessing what they're likely to want next, and then figuring out the coolest way to hit them up - be that email, social media, ads, or in-app messages - all in one go.

What SMART-AI Does	How It Helps
Predict how valuable each customer will be in the future	Spot customers who are about to stop shopping and win them back
Decide where to spend the marketing budget for best results	Detect when a customer is close to making a purchase

Personalize content for each customer in real time	Adjust messaging based on how customers feel about the brand
Coordinate campaigns across email, social media, and ads	Automatically test different versions of campaigns

6.2 How Does SMART-AI Work? (3 Simple Layers)

Layer 1 - Collect All Customer Data So first, SMART- AI pulls up data from everywhere (website, app, emails, socials, even in-store visits) then stitches it into one full profile, per customer. Basically the system always knows the whole picture of who the customer is.

Layer 2 - Make Predictions Using fancy AI models, it guesses stuff like: Is this peep gonna buy soon? Are they about to bail? What kind of content do they get on to? Different AI models take care of different kinds of predictions, all co-existing together.

Layer 3 - Make the Best Decision: Based on those predictions, the system automatically decides the best action to take for example, sending a discount email to a customer who is about to leave, or showing a specific product ad to someone who is likely to buy. All of these decisions can be explained in plain language so managers understand why the AI made each choice.

The Key Feature That Makes SMART-AI Different

Most AI systems are 'black boxes' - they just make decisions and you can't actually see why they do it. SMART-AI employs a technique known as Explainable AI (XAI), which explains to managers the reasons behind each decision in simple and easy-to-understand terms, making it much easier for real people to trust and actually use it.

7. Research Objectives / Hypothesis

7.1 Main Goal of This Research

So, we're trying to build and test this SMART-AI system to prove it really helps retailers bump up their marketing game, more sales, stronger customer relationships, and less money flying out the window, when you compare it to the old school marketing ways.

7.2 Specific Things We Want to Find Out

1. How much better does AI driven marketing really do compared to the old school manual approach that most retailers still use?
2. Is pulling data from multiple sources - like website, email and social media - more accurate in predicting what customers want?
3. Do marketing managers feel more confident and actually want to follow the recommendations of AI when the system explains its reasoning in simple language?
4. Can AI automatically manage and divide a marketing budget among the different channels than a human doing it manually?

7.3 Our Hypotheses (What We Expect to Find)

A hypothesis is simply a prediction that we make before carrying out the experiment. Then we'll see if we really guessed right.

H1: Retailers that use SMART - AI will get significantly more people to finish a purchase than those that stick with old - school marketing methods.

H2: Data from multiple channels will make more accurate (at least 25%) predictions of what customers want in comparison to just one source.

H3: When the AI provides an explanation of its decision-making, marketing managers will feel at least 40% more confident about these decisions than when there's no explanation.

H4: When AI takes care of the marketing budget we expect to obtain at least 20% better return on the money spent compared to a fixed and manually set budget.

8. Methodology (How We Will Test)

8.1 Research Plan

We will run this research over 12 months, split into three clear phases. We will compare the SMART-AI system against the traditional marketing approach used by a real retail partner, measuring the difference in results.

Phase	What We Do	How We Do It	Time
Phase 1	Build the System	Review existing research; collect customer data; train and build the AI models	Months 1-4
Phase 2	Run the Experiment	Test SMART-AI against the old system using real customers from a retail partner	Months 5-9
Phase 3	Analyse the Results	Compare results using statistics; survey marketing managers; write up findings	Months 10-12

8.2 Where Will the Data Come From?

- Real customer data: I'll be working with a mid-size online retailer who will be handing over anonymized records from their shopping data, email campaign results, and website behavior data - nothing personal - data from at least 500,000 customers.
- Public datasets: I'm also looking to leverage public retail datasets for the innovation, which are freely available on sites such as Kaggle and UCI, that will be used for training and testing the AI models before we go live with the experiment.

8.3 How Will We Measure Success?

Business Results We Will Track	Technical Results We Will Track
How many customers actually buy (Conversion Rate)	How accurate the AI's predictions are (F1 Score)
How much money is earned for every dollar spent on ads (ROAS)	How much managers trust the AI's decisions (Survey Score)
How much a customer spends over their lifetime (CLV)	How much the return on investment improves (%)
How many people click on marketing emails (CTR)	How many customers stop shopping less often (Churn Reduction)

9. Expected Outcomes

So, based on what other researchers and companies have found, we expect the following improvements when SMART- AI is used compared to traditional marketing approaches:

What We Measure	Current Average	Expected with SMART-AI
Customer Purchase Rate (Conversion Rate)	2.3% (industry average)	3.5 - 4.1% (up to 78% better)
Return on Every Dollar Spent on Ads (ROAS)	3.2x return	4.8 - 5.5x return (up to 72% better)
Email Click-Through Rate	2.6%	5.1 - 6.3% (up to 142% better)
Customers Stopping Shopping (Churn)	Current level	18-25% fewer customers leaving

Wasted Marketing Spend	Current level	30-40% less money wasted
Manager Trust in AI Decisions	Low (no explanation given)	40%+ more confident (AI explains itself)

10. Business Value

10.1 Direct Money Benefits

If SMART-AI works as expected, retailers will see very real financial benefits:

- Some students ask, "What does this mean?" and I would say as a college student, if 52-78% more folks actually buy, than that's a big deal as far as revenue is concerned. A retailer that is pulling in \$50M a year could end up with \$26 to \$39M additional sales per year.
- Less wasted money: From student budget point of view, less wasted marketing spend, by 30-40%, saves big. If a retailer is spending \$5M on marketing, they could save \$1.5 to \$2M every year - money that could be used for growth, instead.
- As an example, it might be that: - Keeping Customers Longer: Preventing people from churning by 18-25% means every customer is worth more over time. Research shows that by retaining just 5% more customers profits can be increased by 25 - 95% - that's a huge win.
- Staying Ahead Of Competitors The more customers use an AI Powered system, the smarter it becomes. Retailers who get SMART-AI early continue to improve while the competition is left further behind.

10.2 Operational Improvements

In addition to monetary benefits, SMART-AI will make daily operations of marketing much more efficient:

- Faster Decision Making: Rather than waiting weeks to analyse campaign data manually, marketing teams get real-time feedback and the system

adjusts automatically. This means decisions that used to take days are now taking place in seconds.

- Less Manual Work: Tasks such as separating customer groups, addressing which ad to display and adjusting budgets are all automated by the AI. This frees up the marketing team to do some creative and strategic work, rather than repetitive tasks.
- Better Team Collaboration: Since the SMART-AI explains all its decisions in a plain language, both the technical staff (data analysts) and the non-technical staff (marketing managers) can understand and discuss the same information. This helps in breaking down the common wall between data teams and business teams.
- Consistent Performance: Campaigns that are run by humans can have varying levels of quality depending on who is handling them. SMART-AI applies the same logic every time in a consistent way and as a result is more reliable and predictable.

10.3 Long-Term Strategic Benefits

- The Problem: 2016 expert: "Turning Data into Money: Most retailers are sitting on huge amounts of customer data they are not using properly." SMART-AI uses that unused data to create a real advantage in business.
- Grows Without Extra Cost: Once SMART-AI gets initialized, serving 1 million customers will not cost much more than 100,000. This makes it very efficient when the business grows.
- Building AI Skill Inside the Company: Over time, staff get better at what they do with AI and AI gets smarter at what it does with more and more data. Both improve together.

10.4 Benefits for the Whole Industry

If this research is successful, other retailers - big and small - can follow the same approach. This could help the entire retail industry to become smarter and more customer-aware, which is a good thing for businesses and consumers alike.

11. Implementation Plan

Month	Milestone	Key Tasks	Output
1-2	Getting Started	Read existing research; get ethical approval; agree data access with retail partner	<i>Research Plan Document</i>
3-4	Build SMART-AI	Set up data system; build customer profiles; train AI models; test against old baseline	<i>Working SMART-AI System</i>
5-6	Small Test	Run a small experiment to check the system works; make improvements; add explainability	<i>Small Test Report</i>
7-9	Full Experiment	Run the full experiment with real customers; monitor results; adjust as needed	<i>Full Dataset</i>
10-11	Analyse Results	Compare SMART-AI vs old system using statistics; survey managers about trust	<i>Results Report</i>
12	Share Findings	Write the final report; prepare competition presentation; give industry recommendations	<i>Final Proposal & Paper</i>

12. Risks & Ethics

12.1 Possible Risks and How We Will Handle Them

Risk	How We Will Prevent It
Customer data gets leaked (Very serious)	Follow strict data privacy rules (GDPR/CCPA); remove all personal details from data before use
AI treats some customers unfairly (Very serious)	Regularly check the AI for bias; make sure it treats all types of customers equally
Retail partner pulls out of the study (Medium risk)	Have backup agreements with 2 other retailers; use public datasets as a backup
AI model works in tests but fails in real life (Medium risk)	Test the model carefully before going live; keep retraining it with fresh data
Breaking data privacy laws (Very serious)	Get legal approval before starting; design the whole system around privacy from the start

12.2 Ethical Rules We Will Follow

- Customer Consent We will only use customer data with proper permission. Any customer who does not want to have their data used will have the option to opt-out. We are never going to trick or mislead customers.
- Fairness: We will monitor regularly to ensure that the AI is not treating customers differently based on their age, gender, income or background. Everybody deserves to be subject to fair and relevant marketing.
- Only Collect What is Needed) We will only be collecting the minimum amount of data needed to make the system work. We will not collect and store data that is not needed.

- Explainability: Since we will build the AI to explain the decisions in simple language, the managers will always know the reason the system made a specific recommendation. No decision will be totally concealed within a 'black box'.
- Honest Reporting We will honestly report all of our results - including anything that didn't work as expected. Research will only be useful if it is the truth.

13. Conclusion

Retail and E-COMM are changing at lightning fast speeds. Shoppers are now used to personalised interactions and not being just another item in a large and anonymous database. Retailers who can't keep up with this change are losing both customers and revenue to their rivals that make better use of data.

In this study, we present SMART-AI, a purpose-built AI system that has been designed to help retailers better understand their customers, make better marketing decisions, and get more value from the data they are already collecting. Rather than simply adjusting one element of the marketing mix, SMART - AI approaches the whole process: Integrating data across all channels, making correct predictions, and enabling real-time action.

13.1 Why This Research Is Important

This research has importance to two key groups of people. For businesses, it provides a tried and tested blueprint for applying AI to marketing - an area that most retailers are still struggling with. Instead of wasting years and millions of dollars trying to figure it out on their own, they're going to have this clear, evidence-based framework to follow. For academics and students, it represents an important gap in knowledge about AI in marketing. A lot of studies only look at AI tools in isolation. This paper is one of the first to explore how a fully integrated AI system can be used to drive strategic marketing decisions for an entire business and will open doors for research in this fast-growing field in the future.

The potential results speak for themselves - up to 78% more customers making a purchase, up to 40% less money thrown at ineffective ads and up to 25% fewer customers walking away. These numbers are not statistics, they are real business impact that any retailer would want.

Hey, The Most Important Thing I Can Tell You about SMART - AI is That It's Built to be Explainable and Trustworthy. It doesn't just flip the switch on decisions for managers - it actually walks them through the reason in easy to understand language. That's what makes AI a real partner in decision making instead of a stone cold replacement for human judgment.

Final Thought

SMART-AI is not only another marketing hack, but a smarter way to run a business. In this paper I'll walk you through, step by step, how AI can boost a retailer's ability to attract more customers, keep them longer and allocate marketing dollars more efficiently - all by digging into the data that was already there.

References

Chaffey, D. & Ellis-Chadwick, F. (2022). Digital Marketing: Strategy, Implementation and Practice (8th ed.). Pearson Education.

Davenport, T. H. & Ronanki, R. (2018). Artificial Intelligence for the Real World. Harvard Business Review, 96(1), 108-116.

Kietzmann, J., Paschen, J. & Treen, E. (2018). Artificial Intelligence in Advertising. Journal of Advertising Research, 58(3), 263-267.

Kumar, V., Rajan, B., Venkatesan, R. & Lecinski, J. (2019). Understanding the Role of Artificial Intelligence in Personalized Engagement Marketing. California Management Review, 61(4), 135-155.

Li, H. & Kannan, P. K. (2014). Attributing Conversions in a Multichannel Online Marketing Environment. Journal of Marketing Research, 51(1), 40-56.

Lundberg, S. M. & Lee, S. I. (2017). A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems, 30, 4765-4774.

Mikalef, P., Krogstie, J., Pappas, I. O. & Pavlou, P. (2020). Exploring the Relationship Between Big Data Analytics Capability and Competitive Performance. *Information & Management*, 57(2), 103169.

Statista. (2025). Global E-Commerce Sales 2024-2028. Statista Research Department.

Syam, N. & Sharma, A. (2018). Waiting for a Sales Renaissance in the Fourth Industrial Revolution: Machine Learning and Artificial Intelligence in Sales Research and Practice. *Industrial Marketing Management*, 69, 135-146.

Wedel, M. & Kannan, P. K. (2016). Marketing Analytics for Data-Rich Environments. *Journal of Marketing*, 80(6), 97-121.

Huang, M. H. & Rust, R. T. (2021). A Strategic Framework for Artificial Intelligence in Marketing. *Journal of the Academy of Marketing Science*, 49(1), 30-50.

Verhoef, P. C., Broekhuizen, T., Bart, Y., Bhattacharya, A., Qi Dong, J., Fabian, N. & Haenlein, M. (2021). Digital Transformation: A Multidisciplinary Reflection and Research Agenda. *Journal of Business Research*, 122, 889-901.

McKinsey & Company. (2023). The State of AI in 2023: Generative AI's Breakout Year. McKinsey Global Institute Report.

Salesforce Research. (2024). State of Marketing Report (8th ed.). Salesforce Inc.